Design Write-up

Design Process

As a team, we brainstormed project ideas and converged to select one we felt the most excited about, which landed us to create an RPG-style game with characters that we felt represented us as prescribed by different character classes. We then iterated based on this idea (see *figure 1* under *Game Concept Sketches* below), to ideate which features and concepts we wanted in our web application. Following this initial step in the ideation process, we created scenarios (starting from panels 1-8) on visuals of how our application would be laid out, with each scenario capturing a specific feature from our brainstorming or ideation process involving the main gameplay loop emphasizing the interactions between the party members centered around our unique buttons of choice. After the overall layout of the application was designed collectively, we diverged and asked for each individual group member to create a sketch of their own in-game character. We requested that each group member design their character and class based on their unique strengths and weaknesses while also reflecting their personality traits as demonstrated in the ITP personality metrics survey in order to inspire the thought process behind design decision-making. These sketches included having the following features: drawing of character (resemblance of self), clothing, accessories, possible weapon choices, and optionally, traits and a colour palette (see *figures 2-4* under *character sketches* below for each member's respective design process).

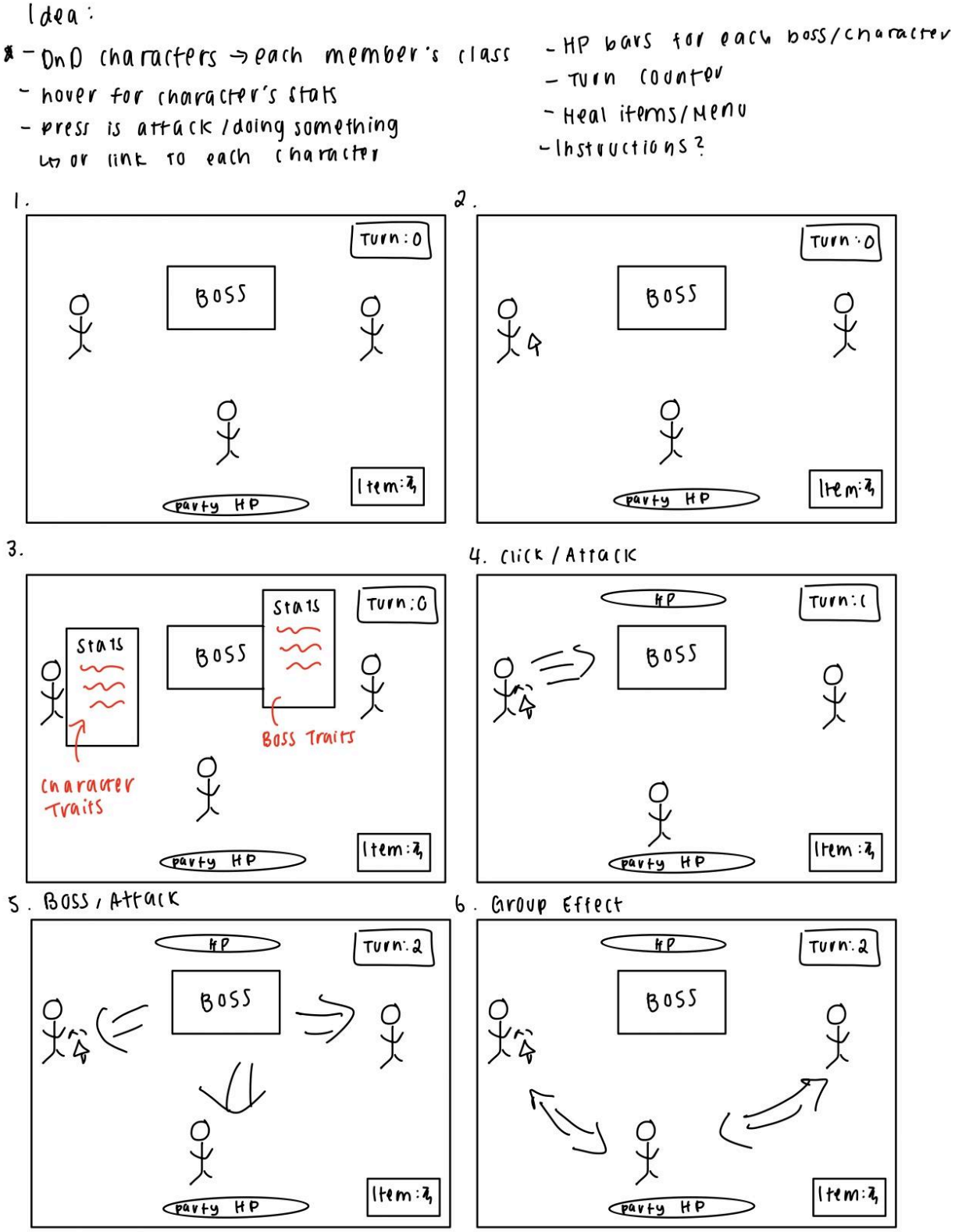
As a whole, we wanted to focus on the special interactions between the members as given by their unique set of traits and selected different in-game stats that would suit each of us as a baseline (e.g., Fion as a marksman having higher attack, Wish as a cleric wizard being lower in attack, and Kevin as a rogue starting with higher defense) while also emphasizing the necessity of teamwork in accounting for each others' weaknesses to create balance within the team. For example, Wish being high in agreeableness, openness in experience, and neuroticism or emotionality scores would translate to taking on a spellcasting role positioned in the backline behind Fion and Kevin as a support player while they take on offense roles being higher in traits like gregariousness and assertiveness. Furthermore, the focus on one button that performs one key action item that serves the purpose of team cohesion while also showcasing each member's distinct quality of traits is demonstrated as a core feature in-game through each member's special

attack. With this feature, we centered on Fion's leadership the attack up button, which makes up for Wish's lower strength as reflected in the lower attack stat, Wish's cooperative nature with the global heal button acting as a form of support for the entire team, and Kevin's ability to plan ahead with the defense up button, which makes up for Fion's higher vulnerability as reflected in her lower defense stat. In this manner, we were able to map out and curate the set of character classes, traits, and qualities that we saw fitting for our own individuality while emphasizing the interplay between the different team members within our group as a cohesive party.

In terms of actual application design, we were able to implement the core feature of button interactivity and turn-taking in consideration of our team as one unit hence why one turn ends after all party members have acted against the singular enemy unit. In addition to this main concept, we also factored in other features that would further enhance the in-game experience such as sound design mapped to characters' attacks and special abilities as a non-visual way to deliver feedback to explore how auditory elements can add to user experience, combat feedback for active communication to make interactions feel more responsive to use, and special items representing our favorite foods as a way to personalize the game for increased enjoyment.

Figures

Game Concept Sketches



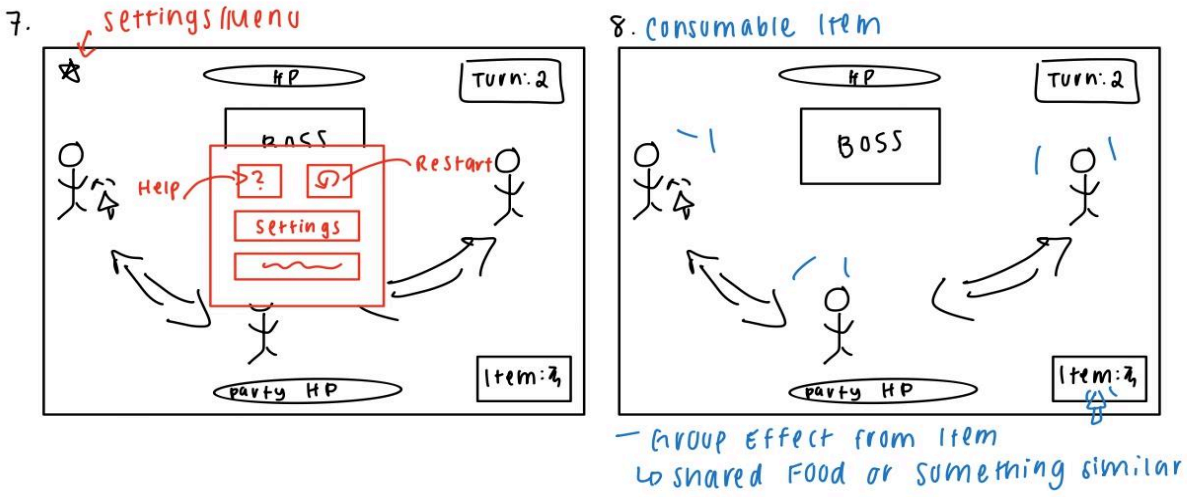


Figure 1: Group brainstorm sketch on our RPG-style game ideations including features we wanted to implement

Character Sketches



Figure 2: Kevin's individual character sketch including possible character animations we need

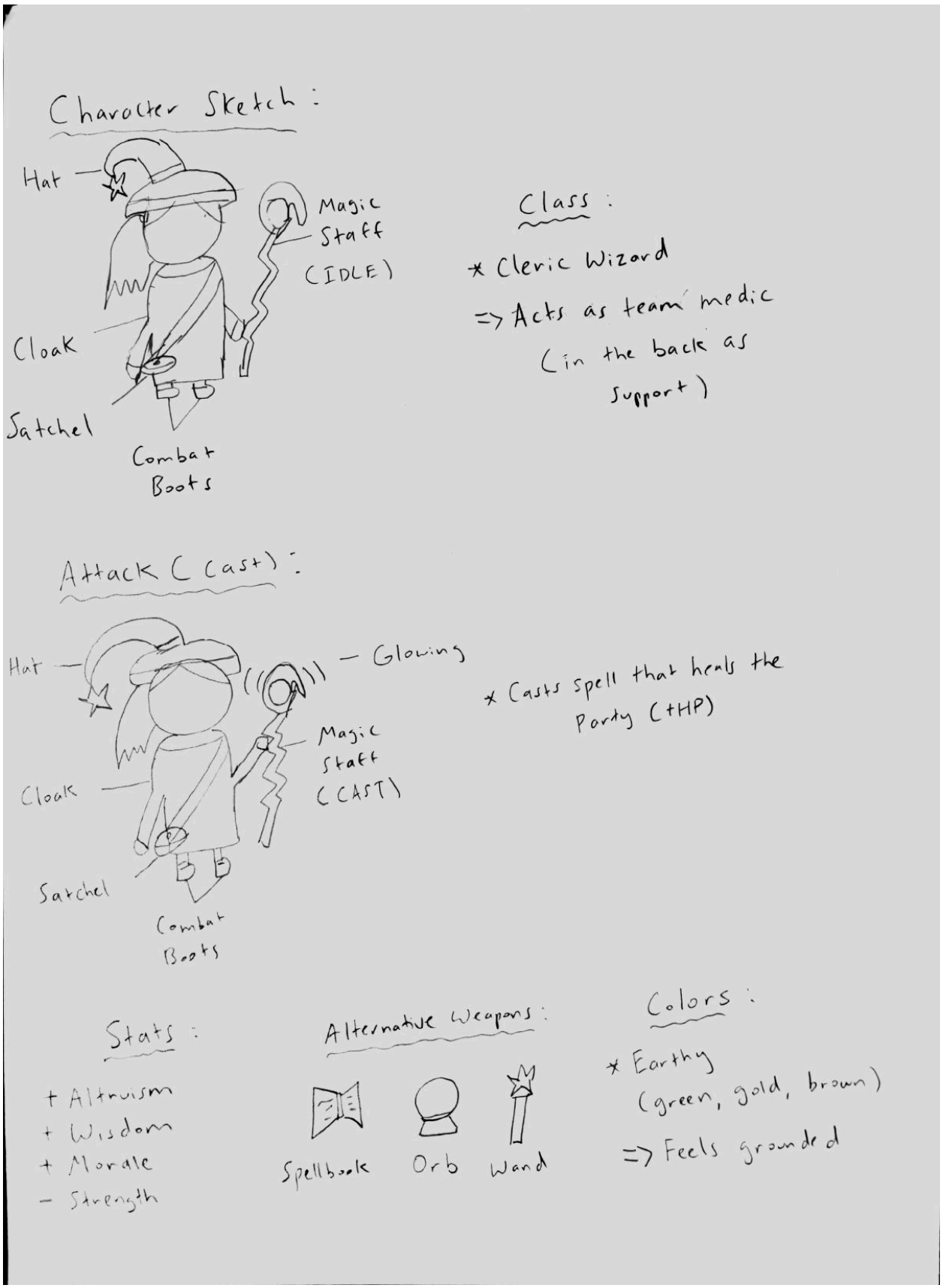


Figure 3: Wish's individual character sketch including weapon attack design

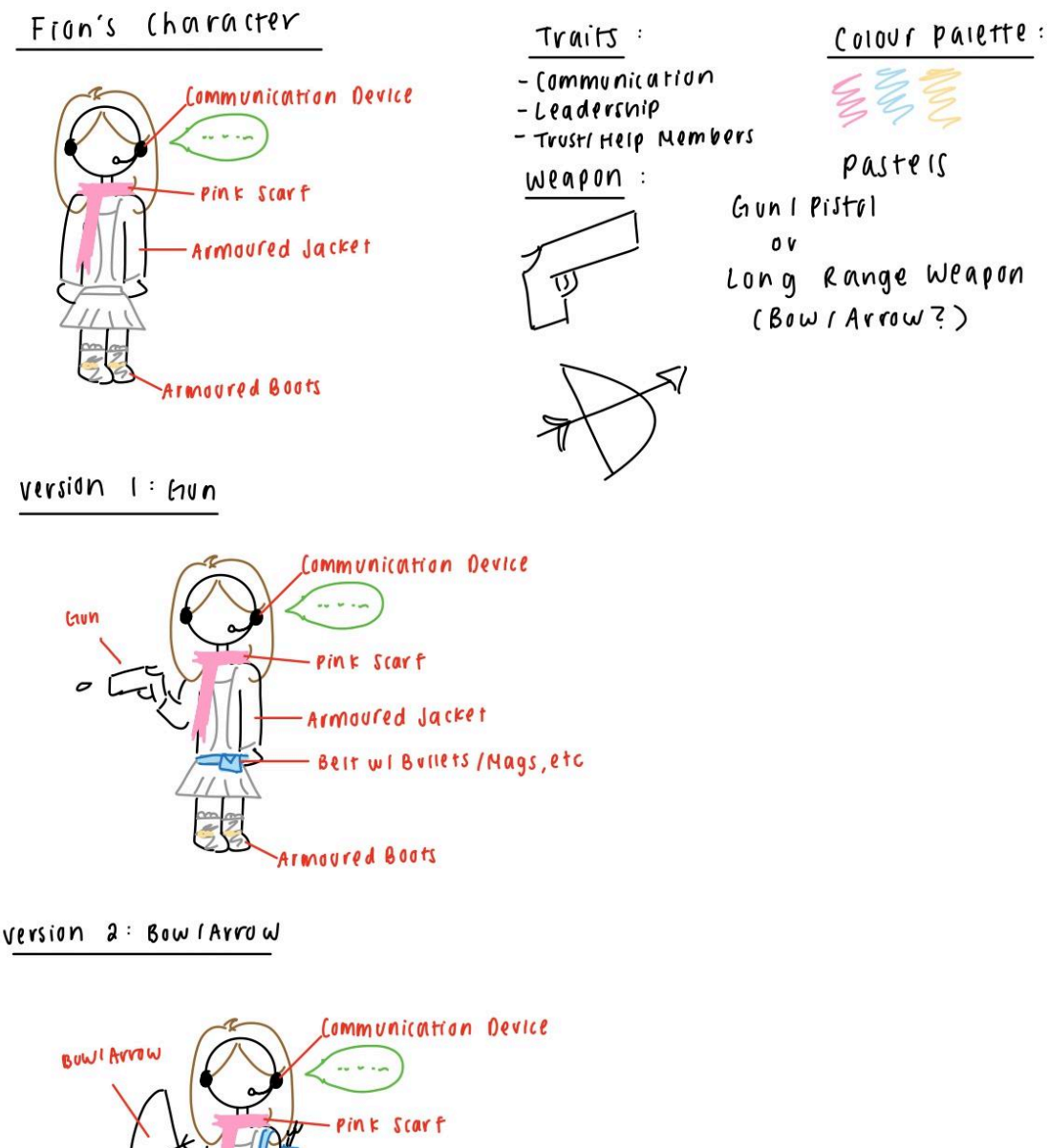


Figure 4: Fion's individual character sketch including alternate design ideas

Functionality

At the time of submission, our core functionality with regards to the web application has been met in terms of showcasing the interactions between characters as well as our individual characteristics with our assigned unique buttons within the overall game. However, if we were to expand the design of our application further, the following pieces of functionality would be implemented and iterated on for enhanced user experience:

- End of battle reward: players will be met with a treasure chest as a reward for defeating the in-game monster, which is helpful for increasing user satisfaction and delight upon completion of an important task.
- Responsive design: the layout of the web application will fit and scale to all screen sizes and aspect ratios, accounting for both desktop and mobile devices for a standardized viewing experience.

Disclosure

Generative AI was used to create all custom characters and spritesheets with animations as seen in our web application. The following prompts were entered into Claude Opus 5.5 to generate materials needed for the project:

- Create a 2D 64x64 pixel-style human character with pale skin, brown eyes, and dark brown hair featuring curtain bangs. Dressed in a baby pink scarf, armored silver jacket and a silver pleated skirt, and armored silver boots adorned with yellow accents. Equipped with a communication device/headset and holding a gun, with a baby blue belt that holds bullets and magazines. This character will have a special group effect, increased attack (use + signs and orange style animation). Characters must have idle, attack, defense, hurt/death (outlined by red for injury), and heal (outlined with green), increased defense, and special attack (outlined with orange) animation different from the other characters.

- Create a 2D 64x64 pixel-style human character with pale skin, brown eyes, and brown-blonde short male style hair. Dressed in black armour with shoulder pads, arcane bracer, utility belt with potions, and brown work boots. Equipped with dark brown

goggles on head and a short sword. This character will have special group effects, increased defense (use shield icons, blue animations). Characters must have idle, attack, defense, hurt/death (outlined by red for injury), heal (outlined with green), increased attack, and special defense (outlined with blue) animations different from the other characters.

- Create a 2D 64x64 pixel-style human character with pale skin, black eyes, and black hair tied in a ponytail. Dressed in a witch cloak, with a satchel, and combat boots. Equipped with a witch hat with a star at the tip, and a magical crooked staff with a bulb/orb on top. When attacking, the orb of the magic staff glows. The colour palette is earthy tones (green, gold, brown). This character will have a special group effect, group heal (use + signs and green aura style animation). Characters must have idle, attack, defense, hurt/death (outlined by red for injury), special heal spell (outlined with green), increased attack, and increased defense animation. This character casts spells that will heal the party so the heal animation can be differentiated amongst the other heal animations

- Each character must output a consistent customization, accessories/clothes, weapon, and colour palette matching its description. Once each character is created, output a sprite sheet with the idle, attack, defense, hurt/death, special group heal, increased attack, and increased defense animations. With each special group effect that is casted by each member, they must have a varying distinct animation aside from the others

- For each pixel character, generate a headshot png of each one using the latest rendition of the character sprite

ChatGPT GPT 5.6-Sol was used to generate a rough draft of the narration script for the project demonstration video. The draft was based on our existing application functionality, character roles, and design decisions. We further revised the wording, timing, and structure of the presentation to reflect our design goals. The prompt entered was as follows:

- My job is to come up with a script and demo video of the application, I'd like help creating that (I attached a rough, bullet form copy of my script along with the rough draft of the write-up and a zip file of the project with this)
    - Bullet points used for script draft creation along with other project files

- Script Draft
    - Character design decisions:
        - Fion:
        - Wish:
        - Kevin:
    - Background/Floor
    - Boss spotlight/aura
    - Sounds:
    - Wish Notes:
    - Button Interactions
    - Character Stats/Class
    - How Team Functions
    - Debug Menu Functionality
    - Fion Notes?:
    - Wish Notes?: